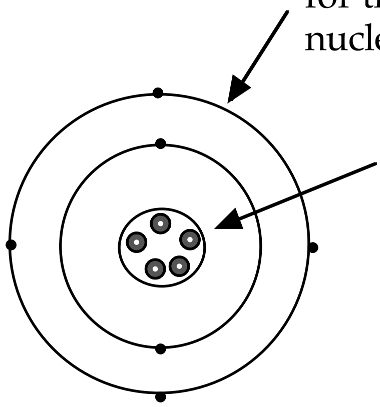
ATOMIC STRUCTURE

Bohr Model of the Atom



Element- contains only one type of atom.

Compound - two or more elements chemically combined in fixed proportions.

Molecule - two or more atoms chemically combined.

Energy Shells - electrons are found in distinct energy shells around the nucleus.

Each of the energy shells has a maximum number of electrons that can be held.

WE WILL CONSIDER THE FIRST 20 ELEMENTS ONLY.

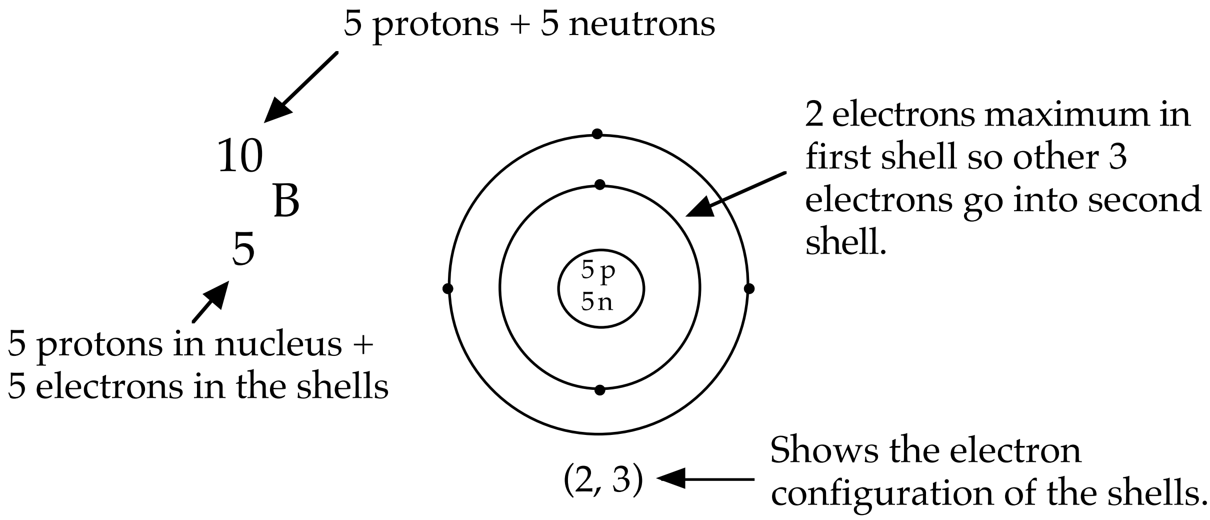
1st shell	2
2nd shell	8
3rd shell	8
4th shell	2

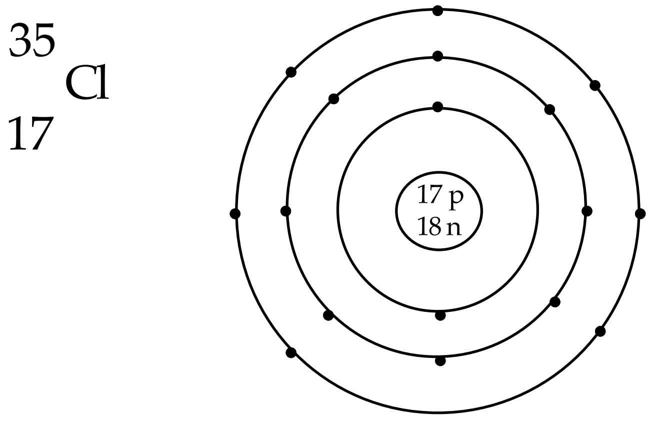
The arrangement of electrons in the shells is called the ***electron configuration*** for the element.

It is possible to draw diagrams to show the electron configuration of elements.

The Periodic Table is very handy for this!

Examples





(2, 8, 7)

FORMATION OF IONS

Definitions

Ion: an atom that has a charge due to the gain or loss of electrons.

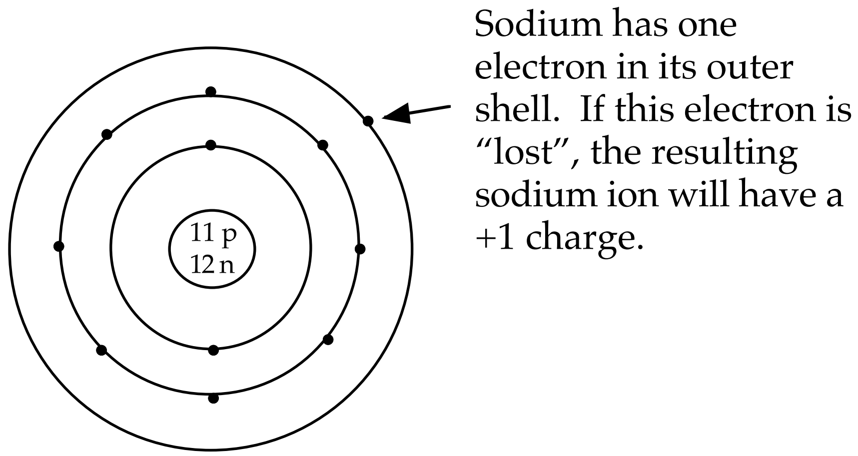
Valency: the charge on an ion.

Positive Ions

- Formed when atoms ***lose electrons***.
- Metal atoms tend to lose electrons from their outer shell.
- Gain a positive charge since there are more protons than electrons.

e.g.

$^{23}_{11}Na$



(2, 8, 1)

When the sodium ion forms, it has a full outer shell of eight electrons like the nearest inert gas - neon - which is very stable.

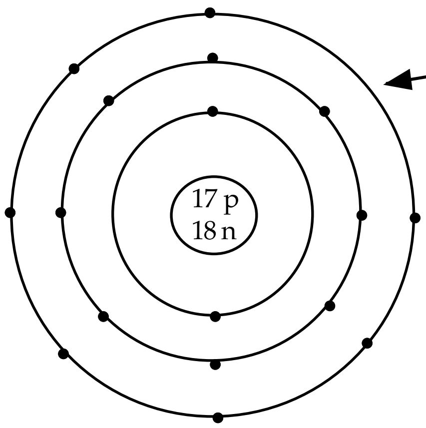
The symbol is written as $\mathbf{Na^{1+}}$.

Negative Ions

- Formed when atoms ***gain electrons***.
- Non-metal atoms tend to gain electrons.
- Gain a negative charge since there are more electrons than protons.

e.g.

$^{35}_{17}Cl$



Chlorine has a space for one electron in its outer shell. If an electron is "gained", the resulting chlorine ion will have a -1 charge.

(2, 8, 7)

By gaining one electron, the chlorine ion has the same completely full outer shell as the nearest inert gas - argon.

The symbol is written as $\mathbf{Cl^{1-}}$.

You will be given a list of common ions that you can use to name and write the formula of various chemicals.

WRITING CHEMICAL FORMULAE

Knowledge of the valency of ions allows us to write formulae to represent different chemicals.

“RULES”

1. Write down the ions involved and look at the valency of each.
2. If the numbers are the ***same***, take one of each ion.
3. If the numbers are ***different***, cross the numbers down to the bottom of each ion.

Examples

magnesium oxide

NUMBERS ARE THE SAME.

Mg^{2+} O^{2-}

TAKE ONE OF EACH ION.

$\therefore$ MgO

barium chloride

NUMBERS ARE DIFFERENT.

CROSS NUMBERS DOWN.

Ba^{2+} Cl^{-}

$\therefore$ $BaCl_2$

THE "1" DOESN'T HAVE TO BE PUT IN.

magnesium phosphate

NUMBERS ARE DIFFERENT.

CROSS NUMBERS DOWN.

Mg^{2+} $(PO_4)^{3-}$

USE BRACKETS FOR IONS WITH MORE THAN ONE ATOM PRESENT.

$\therefore$ $Mg_3(PO_4)_2$

Note

The table of ions will be given to you for use in tests and exams - just know how to use them!